SGLT2 Inhibitor Medicaid Uptake: Research Protocol

State-Level Rurality, Socioeconomic Deprivation, and Medicaid Uptake of SGLT2 Inhibitors *Observational Ecological Panel Study — Conference Protocol (AHA-Ready)*

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1. Title Options

Option A (Equity framing — strongest for AHA): *Rural and Socioeconomically Deprived States Show Lower Medicaid Uptake of SGLT2 Inhibitors After Heart Failure and Kidney Disease Expansion: A National Panel Analysis, 2020–2025*

Option B (Research question framing): *Do Rurality and Deprivation Predict SGLT2 Inhibitor Underutilisation in Medicaid? A 51-State Ecological Panel Study Following Heart Failure and CKD Indication Approvals*

Option C (Punchy/headline-style): *Unequal Adoption: State Rurality and Deprivation as Determinants of SGLT2 Inhibitor Prescribing in US Medicaid, 2015–2025*

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2. Background and Rationale

Why SGLT2 inhibitors?

SGLT2 inhibitors (sodium-glucose cotransporter-2 inhibitors) represent one of the most important therapeutic advances in cardiovascular and renal medicine in recent decades. Beyond their original role in glycaemic control, landmark trials — DAPA-HF (2019), EMPEROR-Reduced (2020), DAPA-CKD (2020), and EMPEROR-Preserved (2021) — established dapagliflozin and empagliflozin as guideline-directed medical therapies for heart failure (HF) and chronic kidney disease (CKD), independent of diabetes status. These indications received formal FDA approval between 2020 and 2022, substantially expanding the eligible patient population.

Despite compelling evidence, real-world uptake of SGLT2 inhibitors remains far below guideline-recommended levels, particularly in populations with limited healthcare access. Guideline-to-practice gaps are widest in patients who stand to benefit most: those with established HF and CKD.

Why Medicaid?

Medicaid insures approximately 90 million Americans and disproportionately covers individuals who are low-income, rural, disabled, or from racial/ethnic minority backgrounds — exactly the populations carrying the highest burden of HF and CKD. Medicaid data therefore provide a unique window into access inequities that would be invisible in commercially insured cohorts. Formulary restrictions, prior authorisation requirements, and prescriber unfamiliarity may further suppress uptake in Medicaid populations relative to private payers.

Why rurality and deprivation?

Rural Americans face worse cardiovascular outcomes, lower rates of specialist access, fewer cardiologists and nephrologists per capita, and greater socioeconomic disadvantage than their urban counterparts. Socioeconomic deprivation compounds these barriers through cost-related non-adherence, lower health literacy, and reduced patient-provider contact. If SGLT2 inhibitor adoption is unevenly distributed across states according to rurality and deprivation, this represents a measurable, modifiable equity gap — and a policy-actionable finding.

Why now?

The period 2020–2025 is the natural experiment window: FDA approvals for HF and CKD indications arrived in a discrete, dateable sequence. If uptake diverged by state rurality or deprivation specifically after these approvals, this implicates structural access barriers rather than differences in underlying disease burden. This cannot be demonstrated with claims or EMR data at the individual patient level using freely available national data — but Medicaid State Drug Utilisation Data (SDUD) make a state-level ecological test feasible, rapid, and policy-relevant.

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3. Research Question

Primary: Did state-level rurality and socioeconomic deprivation independently predict lower Medicaid uptake of SGLT2 inhibitors in the post-HF/CKD-indication era (2020–2025), after adjusting for Medicaid expansion status, state demographics, and temporal trends?

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4. Aims and Objectives

Primary Aim

To examine whether state-level rurality and socioeconomic deprivation are associated with lower Medicaid SGLT2 inhibitor prescribing rates in the period following FDA approvals for heart failure and CKD indications.

Specific Objectives

1. **Describe national trends** in Medicaid SGLT2 inhibitor prescribing per 1,000 enrollees from 2015 (or 2020) to 2025, stratified by era (pre- and post-2020).
2. **Quantify the association** between state rurality quintile and quarterly SGLT2 inhibitor prescribing rate per 1,000 Medicaid enrollees, adjusting for Medicaid expansion status and key covariates.
3. **Quantify the association** between state deprivation (Social Vulnerability Index or poverty rate) and SGLT2 inhibitor prescribing, independently and in combination with rurality.
4. **Test whether uptake divergence by rurality/deprivation widened or narrowed** after the 2020–2022 HF/CKD indication expansions using interaction terms (post-2020 × rurality quintile).
5. **Map state-level variation** in SGLT2 prescribing rates at the most recent available year and identify outlier states — including high-deprivation states with unexpectedly good uptake — to generate hypotheses about enabling factors.

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5. Hypotheses

Primary Hypothesis

States with higher proportions of non-metropolitan population (greater rurality) will have lower quarterly SGLT2 inhibitor prescribing rates per 1,000 Medicaid enrollees in the post-2020 era, after adjustment for Medicaid expansion status, year-quarter fixed effects, and state-level covariates.

Secondary Hypotheses

- **H2:** States with higher Social Vulnerability Index scores (greater deprivation) will have lower SGLT2 prescribing rates, independent of rurality.
- **H3:** The gap in SGLT2 prescribing between rural-predominant and urban-predominant states will have widened after 2020 (i.e., the rurality × post-2020 interaction term will be statistically significant and negative for rural quintiles).

- **H4:** Medicaid non-expansion states will have lower SGLT2 prescribing rates than expansion states, and this association will partially mediate the rurality effect.
- **H5:** Dapagliflozin and empagliflozin — the drugs with FDA-approved HF/CKD indications — will show more pronounced rurality/deprivation gradients than canagliflozin (primarily diabetes-indicated), consistent with indication-driven prescribing barriers rather than formulary access alone.

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6. Study Design

Design: Ecological repeated cross-sectional study with state-quarter as the unit of analysis, structured as a state-year-quarter panel model.

Rationale:

This is the correct design for Medicaid SDUD because:

- SDUD reports prescriptions aggregated at the state level, not patient level. Individual-level analysis is not possible.
- Repeated quarterly observations allow analysis of temporal trends, pre/post comparisons, and panel regression with fixed effects.
- The ecological design is well-established in health services research using aggregate drug utilisation data and is routinely published in high-impact cardiovascular journals.
- The panel structure (51 states × multiple quarters) provides adequate statistical power for adjusted regression despite the modest number of state-level units.

Important caveat to state explicitly: Because the unit of analysis is the state-quarter and not the individual patient, causal inferences cannot be drawn. Ecological correlations reflect aggregate-level associations and may not apply to individual patients (ecological fallacy). This is a limitation that must be acknowledged but does not diminish policy relevance.

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7. Study Period

Primary Analysis Period: Q1 2020 – Q4 2024 (or Q2 2025 if available)

Rationale: This captures:

- Q1 2020 as the pre-dapagliflozin HF-indication baseline (approval: May 2020)
- Full rollout of dapagliflozin HF (2020) and CKD (2021), and empagliflozin HF (~2021–

2022) indications

- COVID-19 pandemic disruption as a confounder to be addressed in sensitivity analysis
- At least 4–5 post-indication years for trend analysis

Extended Analysis: Q1 2015 – Q4 2019 as Pre-Expansion Baseline

Including 2015–2019 enables:

- Characterisation of baseline prescribing during the diabetes-indication-only era
- A formal pre/post interrupted time series analysis with the 2020 indication as the policy shock
- Assessment of whether rurality/deprivation gaps in SGLT2 prescribing existed pre-2020 (which would suggest formulary/cost barriers) or emerged after 2020 (which would suggest indication-adoption barriers)

Recommendation: Run the primary analysis over 2020–2024/25, with 2015–2019 as a pre-period comparison arm in the interrupted time series sensitivity analysis. Present 2015–2025 national trend figure in the main manuscript.

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8. Population and Unit of Analysis

Unit of analysis: State-quarter observation (e.g., Alabama, Q2 2021)

Spatial coverage: All 50 US states plus the District of Columbia (n = 51). Territories (Puerto Rico, US Virgin Islands, etc.) should be excluded from primary analysis given differential Medicaid structures, but may be included in a sensitivity analysis.

Denominator: Medicaid enrollees per state per quarter (or per year if quarterly enrollment unavailable), obtained from the Medicaid Budget and Expenditure System (MBES) or CMS Medicaid enrollment reports.

Note: SDUD data are at the state level by product and quarter. There is no patient-level, county-level, or diagnosis-level information. The study cannot identify individual patients, clinical indications, or prescriber characteristics.

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9. Data Sources

Data Element	Source	Notes

SGLT2 prescribing (numerator)	CMS Medicaid SDUD	Quarterly; state, NDC, drug name, prescriptions, units, reimbursement
Medicaid enrollment (denominator)	CMS MBES / Medicaid enrollment reports	Quarterly or annual; used to calculate prescriptions/1,000 enrollees
Rurality (primary exposure)	USDA Economic Research Service — Rural-Urban Continuum Codes (RUCC), aggregated from county to state	2013 vintage; update to 2023 if released
Deprivation (secondary exposure)	CDC/ATSDR Social Vulnerability Index (SVI)	2020 or nearest available; overall SVI or theme-specific subscores
Medicaid expansion status	KFF State Health Facts	Time-varying variable by state and year
Diabetes prevalence	CDC BRFSS / CDC Diabetes Surveillance System	State-year; adjust for baseline diabetes burden
HF/CKD prevalence	CMS Medicare prevalence data (proxy); BRFSS	Imperfect but available state-level estimates
Clinician density	HRSA Area Health Resources Files (AHRF)	Cardiologist/nephrologist per 100,000 population
Age structure	US Census / ACS	% population ≥65 years by state-year
Race/ethnicity	US Census / ACS	% Black, Hispanic by state-year
Poverty rate	ACS 5-year estimates	Supplementary to SVI
Median household income	ACS 5-year estimates	Supplementary covariate

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10. Drug Classification

Identifying SGLT2 Inhibitors in SDUD

SDUD contains National Drug Codes (NDCs) and drug name fields. A reproducible classification strategy is essential.

Step 1: Primary drug list (HF/CKD indication focus)

- Dapagliflozin (Farxiga; AstraZeneca)
- Empagliflozin (Jardiance; Boehringer Ingelheim / Eli Lilly)

Step 2: All-class secondary list

- Canagliflozin (Invokana; Janssen)
- Ertugliflozin (Steglatro; Merck) — small market share; include in sensitivity analysis

Step 3: Combination products Include combination tablets but classify and report separately, then include in sensitivity analysis:

- Dapagliflozin + metformin (Xigduo XR)
- Empagliflozin + metformin (Synjardy)
- Empagliflozin + linagliptin (Glyxambi)
- Canagliflozin + metformin (Invokamet)

For primary analysis: **single-agent SGLT2 inhibitors only.** Combination products in secondary/sensitivity analysis.

Step 4: NDC identification method

1. Download the FDA NDC Directory (fda.gov/drugs/drug-approvals-and-databases/ndc-directory).
2. Search for active ingredient = “dapagliflozin” OR “empagliflozin” OR “canagliflozin” OR “ertugliflozin”.
3. Extract all matching 11-digit NDCs.
4. Cross-reference with RxNorm (rxnav.nlm.nih.gov) to confirm drug class and ingredient mapping.
5. In SDUD, filter rows where NDC11 matches the SGLT2 inhibitor NDC list OR where the `drug_name` field contains “DAPAGLIFLOZIN”, “FARXIGA”, “EMPAGLIFLOZIN”, “JARDIANCE”, “CANAGLIFLOZIN”, “INVOKANA”, “ERTUGLIFLOZIN”, “STEGLATRO” (case-insensitive string match as validation).
6. Flag any NDC matches that do not have a corresponding name match (and vice versa)

for manual review.

Step 5: Quality checks

- Confirm no SGLT1-selective inhibitors are included (sotagliflozin).
- Confirm insulin or GLP-1 agonist NDCs are not inadvertently captured.
- Verify total national prescription counts against published IMS/IQVIA estimates or published literature for plausibility.
- Produce and publish as supplementary material a complete SGLT2 inhibitor NDC dictionary with drug name, formulation, strength, manufacturer, and date first appearing in SDUD.

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11. Outcomes

Primary Outcome

Quarterly SGLT2 inhibitor prescriptions per 1,000 Medicaid enrollees, by state and quarter.

Formula: (Total SGLT2 prescriptions in state *s*, quarter *t*) / (Medicaid enrollees in state *s*, quarter *t*) × 1,000

Secondary Outcomes

1. **Dapagliflozin prescriptions per 1,000 enrollees** (by state-quarter)
2. **Empagliflozin prescriptions per 1,000 enrollees** (by state-quarter)
3. **SGLT2 inhibitor share of glucose-lowering medications:** SGLT2 prescriptions as % of (SGLT2 + GLP-1 agonist + DPP-4 inhibitor + sulfonylurea + metformin) prescriptions per state-quarter — a measure of formulary positioning and prescriber preference.
4. **Absolute growth rate:** Percentage change in SGLT2 prescriptions/1,000 enrollees from 2019 baseline to 2024 (latest full year), by state.
5. **Reimbursed amount per 1,000 enrollees** — as a proxy for access intensity (recognising that gross reimbursement does not equal net cost after rebates).
6. **Rural–urban prescribing gap over time:** Difference in mean prescribing rate between the top and bottom rurality quintiles, by year.
7. **Deprivation gradient over time:** Slope of prescribing rate across SVI quintiles, by year.

12. Exposure Definitions

Primary Exposure: State-Level Rurality

Source: USDA Rural-Urban Continuum Codes (RUCC), county-level, aggregated to state level.

Recommended operationalisation:

Calculate **percentage of state population living in non-metropolitan counties** (RUCC 4–9), using county-level population weights from the US Census.

Primary classification: **Rurality quintiles** (Q1 = most urban to Q5 = most rural), based on cross-state distribution.

Secondary classification: Continuous (% non-metropolitan population) for dose-response testing.

Tertiary classification: Three-category — Urban-predominant (>80% metropolitan), Mixed (40–80% metropolitan), Rural-predominant (<40% metropolitan) — for simpler descriptive presentation.

Rationale for quintiles: Quintile classification is interpretable, preserves ranking information, allows non-linear trend detection, and is standard in health disparities literature. A continuous measure will be used in sensitivity analyses. Binary rural/urban risks loss of information given the highly variable state-level rurality distribution.

Note on static vs. time-varying exposure: RUCC vintages are released decennially (2003, 2013, 2023). Apply the 2013 vintage to the 2015–2020 period and the 2023 vintage (if released) to 2021–2025. In practice, state-level rurality rankings change little between vintages; a sensitivity analysis treating rurality as fixed (2013 vintage throughout) is appropriate.

Secondary Exposure: State-Level Socioeconomic Deprivation

Recommended measure: CDC/ATSDR Social Vulnerability Index (SVI)

Why SVI over alternatives:

Measure	Pros	Cons
SVI	Composite, validated, CDC-produced, four thematic domains, publicly	Updated every 2 years; county-level aggregation to state

(recommended)	available	needed
Area Deprivation Index (ADI)	Validated, census-tract level	Designed for census tract; state-level aggregation less intuitive
Poverty rate (ACS)	Simple, transparent, annual	Single dimension; ignores housing, social isolation, minority status
Median household income	Widely understood	Single dimension; inverse scale can confuse

Operationalisation:

- Aggregate county-level SVI scores to state level using population-weighted mean.
- Use the **overall SVI score** (0–1) as the primary deprivation exposure.
- Use individual SVI themes (Socioeconomic Status; Household Composition & Disability; Minority Status & Language; Housing Type & Transportation) in exploratory secondary analyses.
- Classify into **SVI quintiles** for comparability with rurality analysis.
- Rurality and SVI will inevitably be correlated; test for multicollinearity (VIF) and consider joint exposure categories (e.g., high-rurality/high-deprivation vs. low-rurality/low-deprivation) in sensitivity analysis.

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13. Main Statistical Analysis

13a. Descriptive Analysis

1. Calculate mean ± SD (or median + IQR for skewed data) of SGLT2 prescribing rates per 1,000 enrollees by state, year, and rurality quintile.
2. Plot national quarterly trend line (all SGLT2 inhibitors combined) from 2015 (or 2020) to latest available quarter. Annotate FDA approval dates as vertical reference lines.
3. Stratify trend by rurality quintile (five lines on a single figure).
4. Produce choropleth maps of SGLT2 prescribing rate by state for the most recent full year.

5. Table 1: State characteristics by rurality quintile at baseline year (2020).

13b. Primary Regression Model

Model type: Negative binomial regression with log(Medicaid enrollees) as offset.

Why negative binomial: Prescription count data are non-negative integers with likely overdispersion (variance > mean). Negative binomial is preferred over Poisson when overdispersion is present. Both use a log-link and log offset, producing incidence rate ratios (IRRs) directly interpretable as rate ratios relative to the referent group.

Outcome: SGLT2 prescription count per state-quarter (with log enrollment as offset).

Primary model specification:

$$\log(\text{SGLT2 count}_{st}) = \beta_0 + \beta_1 \cdot \text{RuralityQuintile}_s + \beta_2 \cdot \text{SVI}_s + \beta_3 \cdot \text{MedicaidExpansion}_{st} + \beta_4 \cdot \text{DiabetesPrevalence}_{st} + \beta_5 \cdot \text{PropOver65}_{st} + \beta_6 \cdot \text{PropBlack}_{st} + \beta_7 \cdot \text{CardiologistDensity}_s + \alpha_s \text{ (state fixed effects)} + \gamma_t \text{ (year-quarter fixed effects)} + \text{offset}[\log(\text{Enrollees}_{st})] + \varepsilon_{st}$$

State fixed effects (α_s): Absorb time-invariant unmeasured state-level confounders (geography, political context, historical prescribing culture). When state fixed effects are included, the main rurality effect (which is largely time-invariant) will be absorbed; this is expected and correct. In this case, rurality acts through interactions with time.

Year-quarter fixed effects (γ_t): Absorb national secular trends in SGLT2 prescribing common to all states (e.g., post-FDA approval national prescribing uptick, COVID disruptions).

13c. Testing Uptake Divergence After 2020

To test whether the rurality gap widened after FDA HF/CKD approvals, add an interaction term:

$$+ \beta_8 \cdot (\text{PostApproval}_t \times \text{RuralityQuintile}_s)$$

Where PostApproval = 0 before Q2 2020, 1 from Q2 2020 onwards.

A significant negative β8 for higher rurality quintiles would indicate that rurality-associated underutilisation intensified after HF/CKD indication expansion — the primary novel finding.

Alternatively, include **year × rurality quintile interaction** to allow flexible non-linear divergence detection across the full time series.

13d. Handling State-Level Clustering

Standard errors must account for within-state correlation across quarters. Use **cluster-robust standard errors at the state level** (51 clusters). Given relatively few clusters, also test with bootstrapped standard errors.

13e. Reporting

Report IRRs with 95% confidence intervals and p-values. Apply Bonferroni correction or report false discovery rate (FDR)-adjusted q-values for multiple secondary outcomes.

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14. Alternative Analysis Options

A. Simple Descriptive Rural vs. Urban Analysis

- **Method:** Compare mean prescribing rates across rurality quintiles using unadjusted ANOVA or Kruskal-Wallis. Visualise with boxplots or caterpillar plots.
- **Strength:** Fast, transparent, immediately interpretable.
- **Weakness:** No adjustment for confounders; does not account for Medicaid expansion or temporal trends.
- **Best for:** AHA conference poster descriptive section; general audience communication.

B. Negative Binomial State-Quarter Panel Model (Recommended Primary)

- **Method:** As described in Section 13b.
- **Strength:** Accounts for overdispersion; models count data correctly; handles offset; allows covariate adjustment and interaction terms; standard for drug utilisation panel data.
- **Weakness:** Ecological design; state fixed effects absorb main rurality effect; requires careful interpretation.
- **Best for:** Primary analysis in both AHA abstract and journal manuscript.

C. Interrupted Time Series (ITS) / Era-Based Analysis

- **Method:** Segment time series at Q2 2020 (dapagliflozin HF approval). Model pre-trend slope, change in level at interruption, and post-interruption slope, separately for rurality quintiles. Use ARIMA or segmented regression with autocorrelation correction.
- **Strength:** Explicitly leverages the FDA approval as a natural experiment; most causally

motivated design.

- **Weakness:** Requires sufficient pre-period data (≥2015); assumes no concurrent policy changes affect rurality groups differentially; COVID pandemic coincides with 2020 interruption point, creating confounding.
- **Best for:** Journal manuscript as primary or secondary analysis; highlights causal interpretation most explicitly.

D. State Fixed-Effect Panel Model (OLS/TWFE)

- **Method:** Two-way fixed effects (state + year-quarter) OLS on log-transformed prescribing rate. Equivalent to a difference-in-differences estimator if rurality is treated as a binary exposure.
- **Strength:** Standard in health economics; exploits within-state variation over time.
- **Weakness:** When rurality is time-invariant, its main effect is collinear with state fixed effects and cannot be estimated; only the interaction with time is identified.
- **Best for:** Robustness check; most appropriate when time-varying rurality data are available.

Bottom line:

- **AHA abstract:** Use Option B (negative binomial) as primary, Option A as descriptive support.
- **Journal manuscript:** Use Option B as primary, Option C as secondary analysis, Option D as robustness check.

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15. Sensitivity Analyses

List the following in a pre-specified sensitivity analysis plan:

1. **Annual instead of quarterly aggregation** — reduces noise but loses temporal granularity; use if quarterly enrollment data are incomplete.
2. **Excluding suppressed/low-volume states** — SDUD suppresses counts <11 for privacy; exclude states with >20% of quarters suppressed and compare with full-sample results.
3. **Excluding 2020–2021 (COVID disruption)** — removes the two years of maximum healthcare disruption; test whether results hold in 2022–2025 only.
4. **COVID interaction term** — instead of exclusion, add a COVID indicator (2020 Q1–2021

Q4) and interact with rurality to test whether pandemic-era disruption differentially affected rural states.

5. **Dapagliflozin and empagliflozin only** (excluding canagliflozin and ertugliflozin) — sharpens the HF/CKD indication focus.
6. **All four SGLT2 inhibitors combined** — maximises prescription counts; trades indication specificity for precision.
7. **SGLT2 share of diabetes drug class** — use SGLT2 prescriptions as a proportion of all glucose-lowering agents as the outcome, adjusting for differences in diabetes burden across states.
8. **Rurality as continuous** — replace quintile categories with % non-metropolitan population as a linear term; test for non-linear (quadratic) relationship.
9. **Alternative deprivation measures** — replace SVI with poverty rate (ACS), ADI, or median household income; compare direction and magnitude of associations.
10. **Medicaid expansion stratification** — run primary model separately in expansion vs. non-expansion states; test whether rurality effect is modified by expansion status.
11. **Excluding DC** — DC is urban by definition and a clear outlier; test sensitivity to its inclusion.

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16. Missing Data and Suppression

SDUD Suppressed Counts

SDUD suppresses state-drug-quarter cells where prescription counts are between 1 and 10 (reported as "0" or blank) to protect patient confidentiality. This is particularly likely for SGLT2 inhibitors in smaller or rural states in early years.

Strategy:

- Treat suppressed cells as missing, not as true zero prescriptions.
- Impute suppressed cells using midpoint substitution (value = 5) in a sensitivity analysis.
- Alternatively, aggregate to annual data, which reduces suppression frequency.
- Report the proportion of state-quarters with suppressed data by rurality quintile (suppression itself may be informative and disproportionately affect rural states).

Missing Enrollment Denominators

MBES enrollment data may have lags or gaps.

Strategy:

- Use most recently available quarterly or annual enrollment data.
- Interpolate linearly between available annual figures if quarterly data are missing.
- Exclude state-quarters with missing enrollment from primary analysis; include in sensitivity analysis with imputed denominators.

Missing Covariates

Some covariate data (e.g., cardiologist density, HF prevalence) may have gaps by year.

Strategy:

- Use the most recent available year's data and treat as time-invariant if annual variation is minimal.
- Use complete-case analysis for primary model; compare with multiple imputation in sensitivity analysis.
- Report missingness in the data flow diagram.

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17. Bias and Limitations

This section must appear in all outputs and should be framed honestly but not apologetically.

1. Ecological design / ecological fallacy All analyses are at the state level. An observed association between state rurality and SGLT2 prescribing rates cannot be attributed to individual patients in those states being rural. The association may reflect prescriber characteristics, formulary structures, or unobserved state-level policies rather than patient rurality per se.

2. No patient-level diagnosis or indication SDUD contains no diagnostic information. It is impossible to distinguish SGLT2 inhibitors prescribed for type 2 diabetes from those prescribed for heart failure or CKD. The study can only examine total SGLT2 prescribing rates. Any framing must not claim to measure "HF prescribing" or "CKD prescribing" — only total Medicaid SGLT2 prescribing volume.

3. Cannot confirm indication-level adoption The era-based design (pre/post 2020) provides indirect inference about HF/CKD indication adoption, but this inference is ecological and circumstantial. Temporal coincidence with FDA approvals supports but does

not prove causation.

4. Formulary and prior authorisation confounding State Medicaid programmes have different formularies and prior authorisation requirements for SGLT2 inhibitors. These policies may directly drive prescribing variation and may correlate with rurality or deprivation independently. Formulary data are not available in SDUD and cannot be controlled for.

5. State-level rurality masks within-state heterogeneity A state classified as "rural" may contain large urban centres. The exposure is a population-weighted average and cannot capture within-state variation. This biases effect estimates towards the null.

6. Prescriptions ≠ adherence SDUD counts prescriptions dispensed, not filled by patients or taken. Adherence, refills, and therapeutic persistence cannot be assessed.

7. Reimbursement ≠ net cost Gross Medicaid reimbursement in SDUD does not reflect post-rebate net cost, which is commercial-in-confidence and not publicly available. Reimbursement analyses should be interpreted with this caveat.

8. COVID-19 confounding The 2020 indication approval coincides with a major healthcare disruption. COVID-19 may have differentially suppressed prescribing in rural and deprived states for reasons unrelated to SGLT2 indication adoption.

9. Medicaid expansion as a confounder/mediator Expansion status is both a confounder and potentially a mediator of the rurality-prescribing association. Adjustment vs. mediation analysis should be considered depending on the causal model.

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18. Ethics

This study uses publicly available, aggregate, de-identified administrative data published by CMS (SDUD) and other federal agencies. No individual patient data are accessed. No patient recruitment, intervention, or data linkage is involved.

Classification: This study is very likely to be classified as **exempt from full IRB review** under 45 CFR 46.104(d)(4) (research involving publicly available data with no identifiable private information). However, institutional policies vary.

Recommendation: Submit a brief exemption determination request to your institutional IRB/Research Ethics Board. This is standard practice, takes minimal time, and provides institutional cover for publication. Document the exemption determination in the manuscript methods section.

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19. Expected Tables and Figures

Main Tables

- **Table 1:** State characteristics at baseline (2020) by rurality quintile (Q1–Q5). Include: n states, % Medicaid expansion, SVI score, % uninsured, diabetes prevalence, median household income, cardiologist density, SGLT2 prescribing rate at baseline. Present means ± SD or medians (IQR).
- **Table 2:** Adjusted incidence rate ratios (IRRs) from the negative binomial model for rurality quintile (reference: Q1, most urban) and SVI quintile, with 95% CIs and p-values. Present separately for total SGLT2, dapagliflozin, and empagliflozin.
- **Table 3:** Interaction terms (post-2020 × rurality quintile): IRRs and 95% CIs. Interpretation: the degree to which each rurality quintile's prescribing changed differentially after the 2020 HF indication approval.

Main Figures

- **Figure 1:** National quarterly trend in SGLT2 inhibitor prescriptions per 1,000 Medicaid enrollees, 2015–2025. Single line plot with 95% CI shading. Annotate: 2016 (empagliflozin CVD indication), 2020 (dapagliflozin HF indication), 2021 (dapagliflozin CKD), 2021–2022 (empagliflozin HF).
- **Figure 2:** Trends stratified by rurality quintile (five lines, colour-coded). Panel A: absolute prescribing rate. Panel B: relative prescribing rate (indexed to Q1 2020 = 100). Include FDA approval annotation lines.
- **Figure 3:** US choropleth maps (two panels): SGLT2 prescribing rate per 1,000 enrollees in 2019 (pre-expansion) and 2024 (post-expansion). Highlight quintile-based colour scale. Optional inset scatterplot: rurality vs. prescribing rate by state.
- **Figure 4:** Forest plot of adjusted IRRs (from Table 2) for rurality and SVI quintiles with 95% CIs, referencing Q1. Use a log-scale x-axis. Separate rows for total SGLT2, dapagliflozin, empagliflozin.

Supplementary Material

- **Supplement Table S1:** Full SGLT2 inhibitor NDC dictionary (drug name, formulation, strength, NDC11, manufacturer, first SDUD appearance).
- **Supplement Table S2:** State-level SGLT2 prescribing rates by year and rurality quintile (full data).
- **Supplement Figure S1:** Sensitivity analysis results (alternative deprivation measures,

COVID exclusion, etc.).

- **Supplement Figure S2:** Choropleth maps of SVI and rurality by state.

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20. AHA Abstract Skeleton

(AHA Scientific Sessions abstract format: 600-word limit; structured with headings)

[TITLE] *Rural and Socioeconomically Deprived States Show Lower Medicaid Uptake of SGLT2 Inhibitors After Heart Failure and Kidney Disease Indication Expansion: A National Panel Analysis*

Background: SGLT2 inhibitors received FDA approval for heart failure (HF) and chronic kidney disease (CKD) indications between 2020 and 2022, substantially expanding their eligible population. Medicaid insures disproportionate numbers of rural and socioeconomically deprived Americans — populations at elevated HF and CKD risk and with limited specialist access. Whether state-level rurality or deprivation predicted lower Medicaid SGLT2 inhibitor uptake after indication expansion is unknown.

Methods: We conducted an ecological repeated cross-sectional analysis using CMS Medicaid State Drug Utilisation Data (SDUD) from [YEAR] to [YEAR], linking to quarterly Medicaid enrollment (MBES), USDA Rural-Urban Continuum Codes (aggregated to state level as % non-metropolitan population), and CDC Social Vulnerability Index (SVI). The unit of analysis was state-quarter (n = 51 × [N quarters] = [N] observations). The primary outcome was quarterly SGLT2 inhibitor prescriptions per 1,000 Medicaid enrollees. We used negative binomial regression with log(enrollees) as offset, adjusting for Medicaid expansion status, diabetes prevalence, proportion aged ≥65 years, cardiologist density, and year-quarter fixed effects, with cluster-robust standard errors at the state level. Rurality and SVI were classified into quintiles. Interaction terms (post-2020 × rurality quintile) tested whether uptake diverged by rurality after HF/CKD indication approvals.

Results: [NATIONAL TREND SENTENCE: Over the study period, national SGLT2 prescribing rose from X.X to X.X per 1,000 Medicaid enrollees, with the most pronounced increase observed after [YEAR].] [RURALITY GRADIENT SENTENCE: States in the highest rurality quintile had X.X prescriptions/1,000 enrollees in [YEAR] vs. X.X in the most urban quintile (p = X.XX).] [ADJUSTED MODEL SENTENCE: In adjusted negative binomial models, states in rurality quintiles Q4 and Q5 had IRRs of X.XX (95% CI X.XX–X.XX) and X.XX (95% CI X.XX–X.XX) relative to Q1 (most urban; reference).] [INTERACTION SENTENCE: The interaction

between rurality quintile and post-2020 era was statistically significant for Q5 (IRR X.XX; 95% CI X.XX–X.XX; p = X.XX), indicating that the rural prescribing deficit widened/did not widen after HF/CKD indication expansion.] [SVI SENTENCE: Higher state SVI (greater deprivation) was independently associated with lower prescribing (IRR per quintile: X.XX; 95% CI X.XX–X.XX; p = X.XX).] Results were consistent across sensitivity analyses excluding COVID-19 years and analysing dapagliflozin and empagliflozin separately.

Conclusions: [RESULTS-DEPENDENT CONCLUSION SENTENCE — see Section 21.] These findings suggest that structural barriers in rural and deprived state Medicaid programmes may limit adoption of evidence-based SGLT2 inhibitor therapy. Targeted policy interventions — including formulary reform, prescriber education, and expanded specialist access — may be needed to ensure equitable uptake of these life-prolonging medications.

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21. Recommended Final Framing

Use this conclusion sentence as a template (adapt to actual findings):

> "These findings suggest that, at the aggregate state level, greater rurality and higher socioeconomic vulnerability are associated with lower Medicaid SGLT2 inhibitor prescribing rates — a pattern that intensified following FDA approval of dapagliflozin and empagliflozin for heart failure and chronic kidney disease indications — though the absence of patient-level diagnosis data precludes direct attribution to specific indication adoption."

What this framing does correctly:

- Anchors the finding at the state level (ecological), not the patient level
- Does not claim to measure HF or CKD prescribing specifically
- Does not claim causation (uses "associated with")
- Acknowledges the key limitation explicitly in the conclusion itself
- Still makes a policy-relevant, actionable point
- Is publishable in a high-impact journal without overclaiming

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Protocol version 1.0 | Prepared for conference submission and manuscript development
Study registration: Consider PROSPERO registration if systematic review component is added; otherwise, consider Open Science Framework (OSF) pre-registration of the analysis plan prior to data access.